

Michael Figurnov

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[GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

I am a Senior Staff Research Scientist at Google DeepMind in London, working on pretraining of Gemini models. Previously, I was a core machine learning researcher behind AlphaFold 2, with contributions to its architecture, training, and predictive accuracy. My earlier research focused on probabilistic machine learning, gradient estimation, and adaptive computation.

Research experience

Google DeepMind , London, UK	2018–present
Senior Staff Research Scientist	2024–present
Staff Research Scientist	2021–2024
Senior Research Scientist	2019–2021
Research Scientist	2018–2019
Selected projects	
Gemini pretraining	2026–present
AlphaFold 3 & AI for biology	2022–2025
Co-author of AlphaFold 3. Research on protein modelling and LLMs for biology.	
AlphaFold 2	2019–2021
- Designed the Evoformer architecture and co-developed the recycling mechanism. - Improved model training and predictive accuracy. - Trained the models used in AlphaFold 2's winning CASP14 submission.	
Architectural contribution highlighted in John Jumper's Nobel lecture (2024).	

Selected publications

Full publication list: [Google Scholar](#).

Accurate structure prediction of biomolecular interactions with AlphaFold 3

Josh Abramson, Jonas Adler, . . . , **Michael Figurnov**, . . . , John M. Jumper
Nature, 2024

Highly accurate protein structure prediction with AlphaFold

John Jumper*, Richard Evans*, Alexander Pritzel*, Tim Green*, **Michael Figurnov***, et al.
Nature, 2021

Monte Carlo Gradient Estimation in Machine Learning

Shakir Mohamed*, Mihaela Rosca*, **Michael Figurnov***, Andriy Mnih*
JMLR, 2020

Implicit Reparameterization Gradients

Michael Figurnov, Shakir Mohamed, Andriy Mnih
NeurIPS, 2018 (spotlight)

Spatially Adaptive Computation Time for Residual Networks

Michael Figurnov, Maxwell D. Collins, Yukun Zhu, Li Zhang, Jonathan Huang, Dmitry Vetrov, Ruslan Salakhutdinov
CVPR, 2017

PerforatedCNNs: Acceleration through Elimination of Redundant Convolutions

Michael Figurnov, Aijan Ibraimova, Dmitry Vetrov, Pushmeet Kohli
NeurIPS, 2016

* Equal contribution.

Education

Higher School of Economics, Moscow, Russia 2013–2019

PhD in Computer Science, Bayesian Methods Research Group.

Dissertation: [Probabilistic method of adaptive computation time in neural networks](#).

Supervised by Dmitry Vetrov.

Moscow State University, Moscow, Russia 2008–2013

Specialist degree (approx. MSc) in Applied Mathematics and Computer Science.

GPA 4.9/5.0; graduated with honours.

Academic service & teaching

Peer review

Reviewer for JMLR, TMLR, NeurIPS, ICML, CVPR, the Symposium on Advances in Approximate Bayesian Inference (AABI), and the Machine Learning for Structural Biology (MLSB) workshop.

Bayesian Methods of Machine Learning seminar 2013–2017

Co-organizer of the weekly seminar.

Moscow State University, Skoltech, Yandex's School of Data Analysis 2014–2015

Teaching assistant for Bayesian Methods of Machine Learning and Graphical Models courses.

Moscow State University 2014–2015

Lead instructor for Machine Learning practical classes.

Selected talks

Protein structure prediction with AlphaFold 2022

NASSMA AI4Science.

Extending the Reparameterization Trick 2018

DeepBayes Summer School.

Internships & research visits

Google Seattle, USA Jul–Oct 2016

Software Engineering Intern, hosted by Li Zhang.

University of Toronto, Canada Sep–Dec 2015

Visiting student, hosted by Ruslan Salakhutdinov.

Microsoft Research Cambridge, UK Sep–Oct 2014

Visiting student, hosted by Pushmeet Kohli.

Awards

2nd place (out of 85), Relevance Prediction Challenge, Yandex 2011

3rd place award, Intel ISEF 2008

Worldwide scientific projects competition for high school students.